⇒ Classification Evaluation :—

(*) confusion matrix :

Positive	TP	FN
Negative	FP	TN

Persition

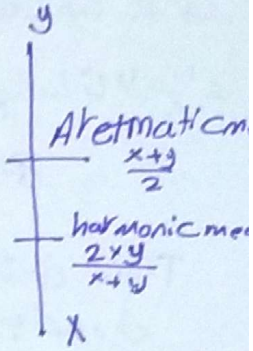
Recall

1- accurecy : $\frac{\text{No. correct predictions}}{\text{No. total data}}$

2- Pecition : $\frac{TP}{TP+FP}$ ما بين كل اللى اتقطو +ve كام واحد فعلا +ve

3- Recall : $\frac{TP}{TP+FN}$ كل اللى فعلا +ve كام واحد اتقط +ve

4- f1 score : $2\,\frac{\text{Percition} * \text{recall}}{\text{Percition} + \text{recall}}$ [harmonic mean]



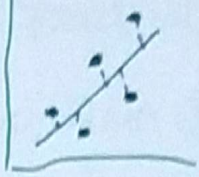
5- R-O-C [Reciver operating characetristics]

TP-Rate = $\frac{TP_s}{\text{All +ves}}$ FP-Rate = $\frac{FP_s}{\text{ALL Negatives}}$

→ Plot each Point and Calculate the area under the curve

Perfect sPlit → 1 Good sPlit → .8 Random sPlit → .5

2 ⇒ Regression Models

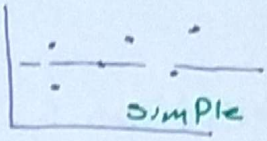


① mean absolute error : absolute of all errors

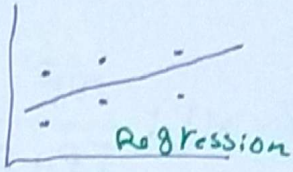
one Problem that it's not diffrentiable so it doesn't work well on the Gradiant Descent for examble

② mean squared error : square of the errors [diffrentiable]

③ R2 score : $1 - \frac{\text{Regression model error}}{\text{simple model error}}$



Bad model $R2 \simeq 0$ / good model $R2 \simeq 1$



⇒ k Fold

⇐ عشان منضيعش داتا مهمه من ال train ونعطيها من ال Test فبنقسم الداتا لأجزاء
كل جزء بيتدرب عليه الموديل و بنعمل Test بباقي الأجزاء وبعدين نجمع النتايج

Train data → Cross validation data → Test data
ادرب الموديل — اختار احسن موديل — اعمل الموديل Test

⇒ If we have more than one Feature to desideon

- Grid search cross validation

for Example SVM

C \ kernel	linear	Polynomial
.1		
1		
10		